

Paul Namalomba - Computational Engineer, Full-Stack Software Developer & Materials Scientist

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Profile views 97

Computational Engineer specializing in Finite Element Analysis R&D, Backend and API Engineering, Data Engineering and Data Science, and most recently Web-stack Development.

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About Me!

I'm a passionate Computational Engineer & Full-Stack Software Developer with expertise in Finite Element Analysis R&D for Civil Engineering applications; Data Engineering, Backend and API Development (REST and Custom APIs) for educational systems and a ride-sharing application --- I have extensive experience with **multiple** Backend Languages and Frameworks (.NET, C#, C++, Python), and with **CI/CD tools** such as GitHub Actions, Bitbucket Pipelines, Docker Compose, Vercel, Cloud Deploy with Google Cloud Run, Azure Virtual Machines and Netlify. have experience with **High-Performance Computing (HPC)**, having developed and optimized scientific computing software using **PETSc, OpenMPI, MUMPS, ScaLAPACK**, and other HPC libraries for parallel processing of large-scale simulations. As I am currently.

I currently am working for a startup called ComputeMore Africa as a **Lead Backend Developer & Chief Innovation Officer** on contract basis, building and maintaining robust C# .NET backend services and APIs for their ride-sharing mobile application, with full Google Maps API integration, JWT authentication, and real-time ride tracking using SignalR. Also working as **SESKA Computational Engineer** at University of Cape Town concurrent with my PhD study (in Computational and Applied Mechanics) where I develop advanced material analysis software and as **SEAT Backend Developer** at University of Cape Town on contract basis, building and maintaining robust Django + PostgreSQL applications for educational enrollment systems.

Current Focus!

- **Backend Development:** Building scalable *Django* applications with *PostgreSQL* databases. Building *Python* apps and modules for data processing and analysis. Building *RESTful APIs* for data access and integration with *C#*.
 - **Computational Engineering & Materials Engineering:** Developing finite element analysis tools with *C++* for advanced materials testing - implementing computational mechanics, solid mechanics, finite element methods, and high-performance computing techniques.
 - **Data Engineering:** Creating robust pipelines for scientific data processing and analysis, ensuring data integrity and performance. Database design, management, and optimization.
 - **Machine Learning & Data Science:** Implementing *machine learning*-based clustering algorithms, outlier detection, and regression analysis for scientific data. Implementing **computer vision** for data analytics, and visualization using *Google Generative AI* and *OpenAI* APIs.
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Professional Experience

Chief Innovation Officer & Lead Backend Engineer - ComputeMore Africa Technologies

Current Position ([**Oct 2025 - present**])

- Developed backend services for ride-sharing application
- OpenAPI compliant RESTful API development
- API docs served with Swagger UI, with COMPLETE test coverage and documentation
- Implemented JWT authentication and role-based access control
- Database schema design and optimization for ride and user data
- Collaborated with front-end developers for seamless integration

Key Technologies: C#, .NET 8/9, .NET ASP and EF Core, PostgreSQL, REST APIs, JWT, Swagger, Git, Docker CI/CD

SESKA Computational Engineer

Current Position (Concurrent with PhD studies [**September 2023 - Present**])

- Develop advanced material analysis software in **C++** and **Fortran** for engineering applications
- Create algorithms for LVDT (Linear Variable Differential Transformer) sensor data processing
- Implement sophisticated data filtering and segmentation techniques using high-performance computing libraries
- **Unique Windows Port:** Successfully ported SESKA software to Windows using **MinGW64** - a challenging and fairly unique implementation leveraging PETSc on Windows
- **CI/CD Management:** Handle all continuous integration and deployment pipelines on **Bitbucket**
- Build visualization pipelines for materials testing analysis

Key Technologies: C++, Fortran, PETSc, OpenMPI, OpenBLAS, ScaLAPACK, MUMPS, METIS, MinGW64, Bitbucket CI/CD

SEAT Backend Developer | SEAT Backend Development Consultant

Current Position ([Dec 2021 - Jan 2022, Nov 2024 - July 2025, Nov 2025 - Feb 2026])

- Develop and maintain Django-based enrollment management system
- Design and implement PostgreSQL database schemas
- Build REST APIs for student enrollment and data management
- Ensure data integrity and system scalability
- Collaborate with academics for user requirements and system improvements
- Implement data validation and integrity checks

Key Technologies: Python, Django, PostgreSQL, REST APIs, Git

Technical Skills

System Architecture and Operating Systems

Windows Cygwin WSL Ubuntu MinGW64 MSYS2

System Scripting & Markup Languages

Windows Batch PowerShell Bash Markdown

Back-end Programming Languages

Python C++ C# ASP.NET SignalR Fortran SQL

Front-end and Web Development Languages

JavaScript HTML5 CSS3 LaTeX TypeScript React

Frameworks & Libraries

Django NumPy Pandas RabbitMQ Scikit-learn Matplotlib Seaborn
SciPy Statsmodels Apache Superset Flask Redis Tailwind CSS BeautifulSoup
Requests

APIs, Protocols and Endpoints

REST API HTTP HTTPS Apache Superset Google Generative AI Google Cloud Google Maps
OpenAI

HPC, Parallel Computing & Scientific Computing

PETSc OpenMPI SuperLU OpenBLAS ScaLAPACK MUMPS METIS ParMETIS

Databases & Tools



Containerization, Integration & Deployment



Specializations

- **High-Performance Computing:** C++/Fortran development with PETSc, OpenMPI, parallel computing
 - **Cross-Platform Development:** Unique Windows porting expertise using MinGW64 for scientific software
 - **CI/CD Engineering:** Bitbucket pipeline management and deployment automation
 - **AI Development:** Generative AI text and image generation, computer vision for data analytics using Google Gemini and OpenAI APIs
 - **Data Science & Machine Learning:** Statistical analysis, clustering algorithms, outlier detection
 - **Scientific Computing:** Materials testing analysis, LOWESS smoothing, gradient analysis
 - **System Administration:** Windows, Linux, automation scripting
 - **Data Engineering:** ETL pipelines, data validation, processing optimization
 - **Backend Development:** Django applications, API-access and integration, PostgreSQL databases
 - **Frontend and Web Development:** React applications, TypeScript, modern UI/UX design with Tailwind CSS
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Featured Projects

Ecoride Ride-Sharing App (Android + iOS + Admin Dashboard + Web Services)

Professional Project - Backend & API Development C# .NET-based backend services for a ride-sharing mobile application. PostgreSQL database design and RESTful API implementation.

Key Features:

- User authentication and role management, custom Auth API with JWT
- API documentation with Swagger UI
- API endpoints for ride management with full, hard Google Maps API integration
- SignalR for real-time ride tracking and updates across driver and user clients
- RESTful API design for enrollment management
- PostgreSQL database with optimized schemas
- Data validation and integrity checks
- Scalability with Kubernetes and Docker CI/CD pipelines
- Plan to push up to Google Cloud run + SQL for production deployment

Architectures & Technologies: Windows, Docker, Google Cloud Run **Back-end:** C#, .NET 8/9, ASP.NET Core, EF Core **Database Architecture and APIs:** PostgreSQL, REST APIs, HTTP/HTTPS, TLS/SSL, Swagger, JWT, SignalR

SESKA Finite Element Analysis Software

Professional Project - SESKA A sophisticated finite element analysis software written in C++ and Fortran for advanced materials testing and computational mechanics applications.

Key Features:

- **Material Damage Mechanics:** Advanced algorithms for material failure analysis and damage progression
- **Viscoplasticity Modeling:** Time-dependent plastic deformation analysis for complex materials
- **Composite Mechanics:** Specialized routines for multi-phase and fiber-reinforced materials
- **High-Performance Computing:** Parallel processing with OpenMPI for large-scale simulations
- **Cross-Platform Support:** Successfully ported to Windows using MSVC++ and MinGW64
- **HPC Integration:** Deployed and optimized for UCT's High-Performance Computing cluster

Technical Achievements:

- **Unique Windows Port:** Challenging implementation leveraging PETSc on Windows with MinGW64
- **Full-Stack Development:** Complete pipeline from scientific research to programming to computational optimization
- **CI/CD Management:** Comprehensive version control and deployment on Bitbucket
- **Library Integration:** Advanced use of OpenMPI, PETSc, MUMPS, ScaLAPACK for parallel computation

Architecture & Technologies: Windows, Linux, Cygwin64 **Back-end:** C++, Fortran, **HPC Libraries:** PETSc, OpenMPI, OpenBLAS, ScaLAPACK, MUMPS, METIS, MinGW64 **Frameworks & Tools:** MSVC++, Bitbucket CI/CD

SEAT Enrollment Management System

Professional Project - Backend Development Django-based web application for managing student enrollment and academic data.

Key Features:

- User authentication and role management
- Student enrollment workflows
- RESTful API design for enrollment management
- PostgreSQL database with optimized schemas
- Data validation and integrity checks
- Scalable architecture for educational institutions

Architectures & Technologies: Rocky Linux, Windows **Back-end:** Python, Django **Database Architecture and APIs:** PostgreSQL, REST APIs, HTTP/HTTPS, TLS/SSL, Celery

Personal Projects

Smart Lame Parameter Calculator



A production-ready React web application that **smartly cross-calculates linear isotropic material parameters** from any two independent inputs. Deployed at www.smartlameparameters.space.

Features:

- **Real-time calculation** from any 2 independent elastic parameters (λ , μ , E, K, ν)
- **Unit conversion** between Pa, kPa, MPa, and GPa
- **Parameter locking** to fix values during recalculation
- **Consistency checking** for contradictory inputs
- **Derivation trace** showing which formulas were used
- **Export functionality** - results as JSON or CSV
- **Permalink generation** for sharing calculations
- **Fully accessible** with ARIA attributes and keyboard navigation
- **Responsive design** with modern UI/UX
- **100% TypeScript** with comprehensive type safety
- **80%+ test coverage** with Jest and React Testing Library

Front-End Technologies: TypeScript, React, Vite, Tailwind CSS **Testing:** Jest, React Testing Library

Deployment: Vercel, GitHub Actions CI/CD, Netlify

MSE AI Agents - AI-Driven Data Collection and Analysis for Stock Market Investment



AI-powered data collection and analysis toolkit for stock market investment research using Google Generative AI and OpenAI APIs. Organised to maximise data insights for investment decision-making. Aiming to give the user an interactive AI assistant and DASHBOARD for stock market data analysis.

Features:

- **Data Collection:** Automated retrieval of stock market data from mse.co.mw - Malawi Stock Exchange, one of the top 3 best performing stock exchanges in Africa to USD.
- **Auto-Rate Adjusted Data Scraping:** Intelligent scraping that adjusts for current request rates to avoid server overload primarily.
- **Automated Data Analysis:** In-depth analysis using advanced statistical and machine learning techniques
- **Interactive Dashboard:** User-friendly interface for visualizing and interacting with data insights
- **AI Integration:** Leveraging Google Gemini and OpenAI for enhanced data interpretation and recommendations
- **Investment Insights:** Tailored insights to support informed investment decisions

Datashadric - Data Science Toolkit



A comprehensive Python package for data science, machine learning, statistical analysis, and visualization. Modularly organized for easy use and pip installation.

Features:

- **Machine Learning:** Naive Bayes models, logistic regression, model evaluation
- **Statistical Analysis:** Normality testing, confidence intervals, hypothesis testing
- **Data Manipulation:** Missing value analysis, data cleaning, one-hot encoding
- **Regression Analysis:** OLS modeling with diagnostic checks
- **Visualization:** Box plots, histograms, scatter plots, pairplots
- **AI Integration:** Data insights and visualization analysis using Google Generative AI and OpenAI APIs

Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, SciPy, Statsmodels

IP Masquerade - Privacy Proxy Server



A flexible HTTP/HTTPS proxy server built with Python and Flask for anonymous web browsing and IP masking with advanced privacy features.

Features:

- **IP Masking:** Hide real IP address from target servers
- **Header Sanitization:** Remove privacy-revealing headers automatically
- **User-Agent Rotation:** Automatic rotation to avoid detection
- **Proxy Chaining:** Route traffic through multiple proxy servers
- **Rate Limiting:** Built-in abuse prevention and stability
- **SSL/TLS Support:** Secure encrypted connection handling

Technologies: Python, Flask, HTTP/HTTPS Protocols, SSL/TLS, Network Security

basketPredictor - Stochastic and Regression-Based Predictor



Data science project combining **web scraping**, **regression analysis**, and **visualization** for NBA basketball game prediction.

Features:

- **Web scraping** for real-time NBA data
- **Advanced statistical analysis**
- **Machine learning regression models**
- **Data visualization** and analysis
- **Investment decision support** tools

Technologies: Python, Pandas, Scikit-learn, Matplotlib, Web Scraping

Sequential PDF Merger



A sophisticated **PDF processing tool** with **GUI interface** for merging multiple PDF files with advanced sorting and organization features.

Features:

- **Intelligent file sorting** algorithms
- User-friendly **Tkinter GUI**
- **Batch processing** capabilities
- **PowerShell automation** support and implementation, for producing .exe files for Windows users
- **Cross-platform compatibility:** Works on Windows, Linux, and macOS through the shell as well.

Technologies: Python, PyMuPDF, PyPDF2, Tkinter

A3 to A4 Converter



Automated **document processing tool** for converting scanned A3 pages to A4 format with **intelligent page detection** and splitting.

Features:

- **Automated page size detection**
- **Smart splitting algorithms** based on page size and content-placement
- **Batch processing** for large documents
- **Image processing optimization**

Technologies: Python, PyMuPDF, PyPDF2, Tkinter

System Management Scripts



Collection of **automation scripts** for **Linux**, **Windows**, and **MSYS64** system administration and management.

Features:

- **Cross-platform compatibility**
- **Automated system maintenance**
- **Performance monitoring** tools
- **Development environment setup**

Technologies: Windows Powershell, Powershell 7+, Bourne-Again Shell

Related Guides: You can find my guides and scripts regarding system management and automation in the following categories:

1. **Technical Documentation:** Comprehensive power user guides for backend engineers, platform engineers, security engineers, and SREs covering enterprise-level technologies. DIRECT LINK: [Technical Documentation Repository](#)
2. **Automation Scripts:** Production-ready PowerShell scripts for Windows system management, file operations, GitHub workflows, and development automation. DIRECT LINK: [Automation Scripts Repository](#)

All guides follow a consistent template with quickstart instructions, key concepts, configuration best practices, security considerations, detailed examples, troubleshooting tips, and performance tuning recommendations.

Simultaneous Equations Solver



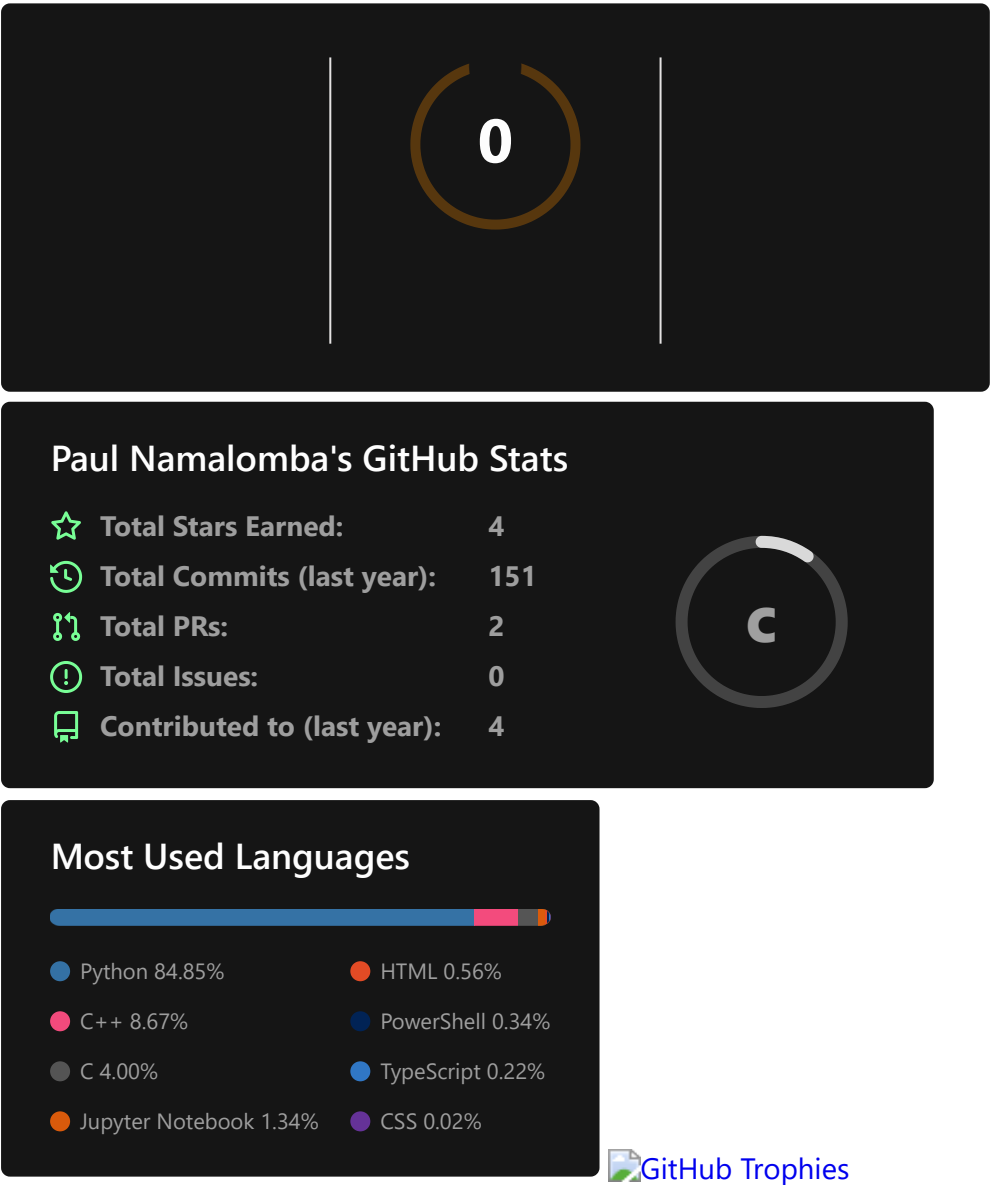
Educational **mathematical tool** for solving systems of **simultaneous equations** with user-friendly input interface.

Features:

- **Multiple equation solving methods** two-variable and three-variable, multiple equation types
- **Step-by-step solution display**
- **Error handling** and validation
- Cross-platform **Mathematics tool** interactable through shell

Technologies: Python Math, NumPy, Linear Algebra

 GitHub Statistics



Technical Expertise Areas

Computational Science

- Materials testing and analysis
- Scientific data processing
- Statistical analysis and modeling
- Algorithm development for engineering applications

Data Engineering

- ETL pipeline design and implementation
- Data validation and quality assurance
- Database design and optimization
- Real-time data processing systems

Software Development

- Full-stack web development
- API design and implementation
- Database architecture
- System integration and automation

Machine Learning & AI

- Clustering algorithms (K-means, DBSCAN)
 - Outlier detection methods
 - Regression analysis and modeling
 - Data visualization and interpretation
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Current Learning & Development

- Advanced machine learning techniques for materials science
 - Cloud computing platforms (AWS, Azure)
 - Microservices architecture
 - DevOps and CI/CD pipelines
 - Advanced PostgreSQL optimization
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Let's Connect

I'm always interested in discussing new opportunities, collaboration on interesting projects, or sharing knowledge about computational science and software development.

Professional Interests:

- Materials science computing
 - Backend and API development
 - Data engineering and analytics
 - Scientific software development
 - Educational technology solutions
 - Open source contributions
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Referees

Prof. Sebastian Skatulla

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Additional Resources

- **Download My Resume (CV)** - Complete professional background and experience - showcasing my skills, certifications and referees!
- **GitHub Profile** - Explore all my projects and contributions, perhaps even collaborate on something new!
- **LinkedIn** - Professional network and updates, and also as a professional instant messaging platform...

"Combining computational science with practical software engineering to solve real-world problems and drive digital innovation."